K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

Lab Assignment : 01 (from Unit -1)

Mini Project Assignment: Foundations of AI in Cyber Security

Course: AI in Cyber Security Lab
Course Code: ENSP355
Assignment Title: End-to-End Mini Project: Building an AI-Driven Cyber Threat Awareness and Detection Prototype
Assignment Type: Individual
Estimated Duration: 12–15 hours
Weightage: 20% of Lab Assessment

Real-World Problem Context

Modern organizations rely heavily on digital infrastructure, making them vulnerable to a wide range of cyber threats such as phishing, malware, insider misuse, and automated attacks. Traditional cyber security tools based on static rules and signatures are often insufficient against intelligent and adaptive threats.

Artificial Intelligence enables cyber security systems to learn patterns from data, identify abnormal behavior, and assist security teams in understanding evolving threat landscapes. Before deploying advanced ML or DL models, it is essential to build a strong practical foundation in AI concepts, threat analysis, and ethical considerations.

This mini project simulates the role of an entry-level cyber security analyst tasked with building a basic AI-enabled cyber security analysis and detection prototype.

Learning Objectives

By completing this lab assignment, students will be able to:

1. Implement basic AI-driven cyber security analysis pipelines
2. Apply foundational Machine Learning concepts to security datasets
3. Analyze and visualize cyber threat patterns using data-driven methods
4. Understand the role of AI in identifying threats and anomalies
5. Reflect on ethical and operational challenges of AI in cyber security

Assignment Tasks

Task 1: Workspace Bootstrap and Environment Setup

• Create a private GitHub repository named:
ai-cybersecurity-unit1-mini-project-<yourname>

• Add the following:

- README.md
- .gitignore
- requirements.txt
- data/ folder for datasets
- src/ folder for Python scripts or notebooks

• Create a Python virtual environment
• Install required packages:

- numpy
- pandas
- scikit-learn
- matplotlib / seaborn
- jupyter

• Validate setup by running a sample Python script or notebook

Task 2: Cyber Security Threat Landscape Analysis (Implementation Task)

• Load a basic cyber security dataset (e.g., login attempts, network traffic summary, phishing indicators)
• Perform exploratory data analysis (EDA):

- Identify common threat indicators
- Visualize frequency of suspicious activities
  • Generate plots showing:
- Normal vs suspicious behavior
- Distribution of attack-related features

Task 3: Basic Machine Learning-Based Threat Classification

• Preprocess the dataset:

- Handle missing values
- Encode categorical features
- Normalize numerical attributes

• Implement a basic supervised ML model:

- Logistic Regression or Decision Tree

• Train the model to classify events as normal or suspicious
• Evaluate performance using:

- Accuracy
- Precision, Recall, F1-score

Task 4: Introduction to Deep Learning Concepts (Prototype Implementation)

• Implement a simple neural network using:

- scikit-learn MLPClassifier or a basic Keras model

• Compare ML vs DL performance on the same dataset
• Analyze:

- Training time
- Detection accuracy
- Practical feasibility

Task 5: Ethical Reflection and AI Challenges (Implementation + Analysis)

• Simulate a scenario where:

- False positives block legitimate users
- Biased data leads to unfair decisions

• Modify thresholds or model parameters to observe impact
• Document:

- Ethical risks
- Privacy concerns
- Explainability limitations

Final Deliverables

Students must submit:

• GitHub repository link containing:

- Python source code (.py or .ipynb)
- README.md with setup and usage instructions
- requirements.txt

• A short technical report (PDF) including:

- Problem overview
- Methodology
- Results and observations
- Ethical discussion

• Screenshots or exported plots of outputs

Evaluation Rubric (Total: 100 Marks)

Criteria	Marks	What is Expected
Repository Setup & Organization	10	Proper structure, clean setup, runnable code
Threat Analysis & Visualization	20	Meaningful EDA and security insights
ML Implementation	25	Correct ML model with proper evaluation
DL Prototype & Comparison	20	Working DL model and comparative analysis
Ethical Analysis & Reflection	15	Clear understanding of AI risks in cyber security
Code Quality & Documentation	10	Clean, commented, well-documented code

Submission Instructions

• Push all project files to the GitHub repository
• Submit the **GitHub repository link** and **PDF report** on LMS
• Ensure code runs without errors
• Late submission penalties apply as per university norms

Academic Integrity & Plagiarism Policy

• This is an individual lab assignment
• Copying code or reports without citation will result in zero marks
• All external datasets and references must be cited

Support Resources

• Scikit-learn Documentation
• Kaggle Cyber Security Datasets
• NIST Cyber Security Framework
• Introductory AI & ML lecture notes (Unit–1)

Deadline: [To be announced]

Faculty: Dr. Amar Saraswat